kind of repulsive energy to it. When you have a lot of space and not much matter (like in the space between stars or galaxies), that repulsive energy can counter gravitational attraction.

Success! The equation worked. It nicely described a static universe in which the existence of other stars or galaxies doesn't immediately collapse the entire cosmos. Einstein had done it again.

Only problem: the universe is not static. This became apparent to the astronomical community a few years later, when it turned out that fuzzy smudges in the sky previously called "spiral nebulae" were actually other galaxies. Soon after, Hubble used the redshifting of those galaxies to show the universe was actually expanding. While a static universe with only attractive gravity is doomed, an expanding universe can be saved, at least temporarily, by its own expansion. The gravity might slow the expansion, and might eventually turn it around, but a universe can get along just fine for many billions of years on an initial growth spurt and the ongoing effects of that expansion. (How the expansion started is a whole other story, but all we need for this particular problem is for the universe to not be so profoundly doomed that it would *already* be toast, and either a cosmological constant or expansion can take care of that.)

The discovery of the expanding universe meant a whole new view of cosmology and a minor embarrassment for Einstein. He somewhat reluctantly removed the cosmological constant term from his equations and wandered off to try to revolutionize some other area of fundamental physics. And so things went, with the evolution of the universe making a reasonable amount of sense, right up until the supernova measurements made a mess of it all again in 1998. Accelerated expansion meant the cosmological constant had to be revived, with the only small mercy being that it was by then far too late for Einstein to say, "I told you so."

Just because a cosmological constant allows the universe to be accelerating in its expansion doesn't mean it's broadly considered to be a good and sensible solution.[fn3] There's nothing that explains why the cosmological constant term should have the value it does, from a theoretical standpoint. Why should it exist at all, except as a suspiciously convenient fix to our equations? And if we have to have a cosmological constant, why not a larger value? One of the most logically natural ways for the universe to have a cosmological constant would be for the constant to derive from the vacuum energy of the universe—the energy of empty space that